FLOOD RISK ASSESSMENT FOR OML 17/22 SEISMIC AREA

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LIST OF ABBREVIATIONS AND ACRONYMS:

DEM	Digital Elevation Model
DTM	Digital Terrain Model
LANDSAT	American Remote Sensing Satellite
FS	Flowstation
OML	Oil Mining Lease
MF	Manifold
Pixel	One picture element or cell in a digital image.
RGB	Red, Green, Blue – primary colours used in digital colour displays
RoW	Right of Way
EGM	Earth Gravitational Model
TMNIGM	Transverse Mercator Nigerian Mid belt

EXECUTIVE SUMMARY

This report is on the flood risk assessment of OML 17/22 seismic area of interest. Landsat 8 satellite imagery, Seismic DEM data, TerraSAR-X DTM, 2012 High water Mark data and 2012 empirical flood data were used to determine the areas that are most susceptible to flooding within the study area. The Seismic DEM data, Land cover classification and High Water Mark values were integrated to predict the areas that are susceptible to flooding, while the 2012 empirical flood data was used as a further validation of the flood susceptible model. Land cover was ranked on a scale of 1 to 5 relating to the susceptibility level of each of the classes based on visual interpretation thereby analysing each land cover class within the flood risk affected areas. Elevation is an important factor to susceptibility determination. Places with elevation values below the High Water Mark of the surrounding river are more susceptible to flooding than area with heights above the High Water Mark. The study identified areas with potential flood risk and it is hoped that it would support in executing decisions and proper planning for the study area.

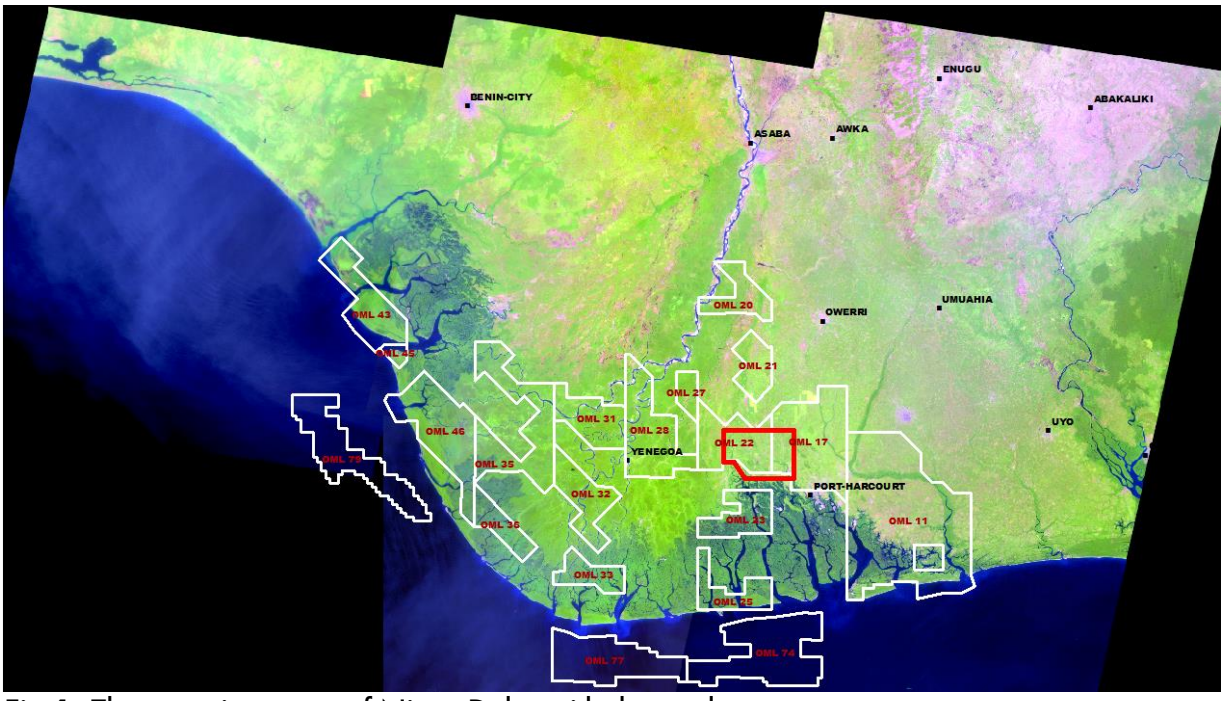

Fig 1: The overview map of Niger Delta with the study area

1.0 INTRODUCTION

The OML 17/22 seismic area of interest under review has an area of approximately 595.5km^2 and has height values between -0.19 and 29.29 metres (Datum: Mean Sea Level). The Land cover class consist of 6.42km^2 (1%) of Water, 88.09km^2 (15%) of Primary forest, 283.48km^2 (48%) of Secondary forest which is the most dominant land cover class within the study area. Farmland vegetation constitutes of 187.09km^2 (31%) and Urban/Industrial class constitutes of 30.42km^2 (5%). Settlements within the urban/industrial class are Obelle, Omegwa, Ovogo, Ihuowo, Isiokpo, Omagwa, Choba, Emuoha, Rumuchie amongst others. It also includes SPDC locations which are Obelle FS, Rumuekpe FS, and Ibaa MF.

2.0 DATASETS USED FOR THE STUDY

The datasets used for this project are Landsat8 imagery acquired on January 2016, Seismic DEM data and TerraSAR-X DTM with 10m resolution. These data are the subsets of the original images kept in the Geomatics archives. The images were geo-referenced in MINNA datum and TMNIGM projection. Other vector data used for the projects are the RoW parcels, SCiN pipelines, Wells e.t.c.

Table1: Data Properties of Landsat8

Landsat 8 Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS) Launched February 11, 2013	Bands	Wavelength (micrometers)	Resolution (meters)
	Band 1 - Coastal aerosol	0.43 - 0.45	30
	Band 2 - Blue	0.45 - 0.51	30
	Band 3 - Green	0.53 - 0.59	30
	Band 4 - Red	0.64 - 0.67	30
	Band 5 - Near Infrared (NIR)	0.85 - 0.88	30
	Band 6 - SWIR 1	1.57 - 1.65	30
	Band 7 - SWIR 2	2.11 - 2.29	30
	Band 8 - Panchromatic	0.50 - 0.68	15
	Band 9 - Cirrus	1.36 - 1.38	30
	Band 10 - Thermal Infrared (TIRS) 1	10.60 - 11.19	100 * (30)
	Band 11 - Thermal Infrared (TIRS) 2	11.50 - 12.51	100 * (30)

3.0 WATER LEVELS

There are no records of water level monitoring within the OML 17/22 study area. The Nun River monitoring data (High Water Mark value) was used as a reference to the surrounding rivers within the OML 17/22 project area since that is the period we had flooding within the study area. The Nun River water level monitoring has a high water

mark of 8.51meter above mean sea level based on 2012 SPDC High Water Mark record.

4.0 LAND COVER CLASSIFICATION

Land cover type of the area contributes to determining its susceptibility to flooding. Places that are within proximity to water bodies are more likely to be flooded than area farther away. Elevation is also an important factor, as areas with elevation values below the High Water Mark of the surrounding river are more susceptible to flooding than area with heights above the High Water Mark.

A Land cover classification was done using an unsupervised classification process, which uses an iterative ISOclass algorithm to generate a number of classes pre-defined by the processor. This classification technique groups similar pixels based on the spectral characteristics of the image datasets. This method does not require any prior knowledge of the land cover types in the area of interest. The following parameters were used for the unsupervised classification process; minimum class of 15 and which was later re-classed to 5 classes.

Water

This comprises mainly the major river and its tributes within the study area.

Primary Forest

This type of forest is an indication that there are no clearly visible indications of human activities and the ecological processes are not significantly disturbed. These vegetation types are mostly found around the water bodies within the study area.

Secondary Forest

This type of forest replaces the primary forest and is the result of vegetation regrowth after loss of primary forest which may have been cleared by natural or man-made causes such as agriculture or urban development.

Farmland

Most of these farmlands are found in close proximity to human settlement within the study area.

Urban/Industrial

This area of interest is predominantly built-up with settlements like Obelle, Omegwa, Ovogo, Ihuowo, Isiokpo, Omagwa, Choba, Emuoha, Rumuchie amongst others.

Table2: Land Cover Classification for 2015

Land Cover Types	Area (km^2)	Percentage cover (%)
Water	6.42	1
Primary Forest	88.09	15
Secondary Forest	283.48	48
Farmland	187.09	31
Urban/Industrial	30.42	5
Total	**595.5**	**100**

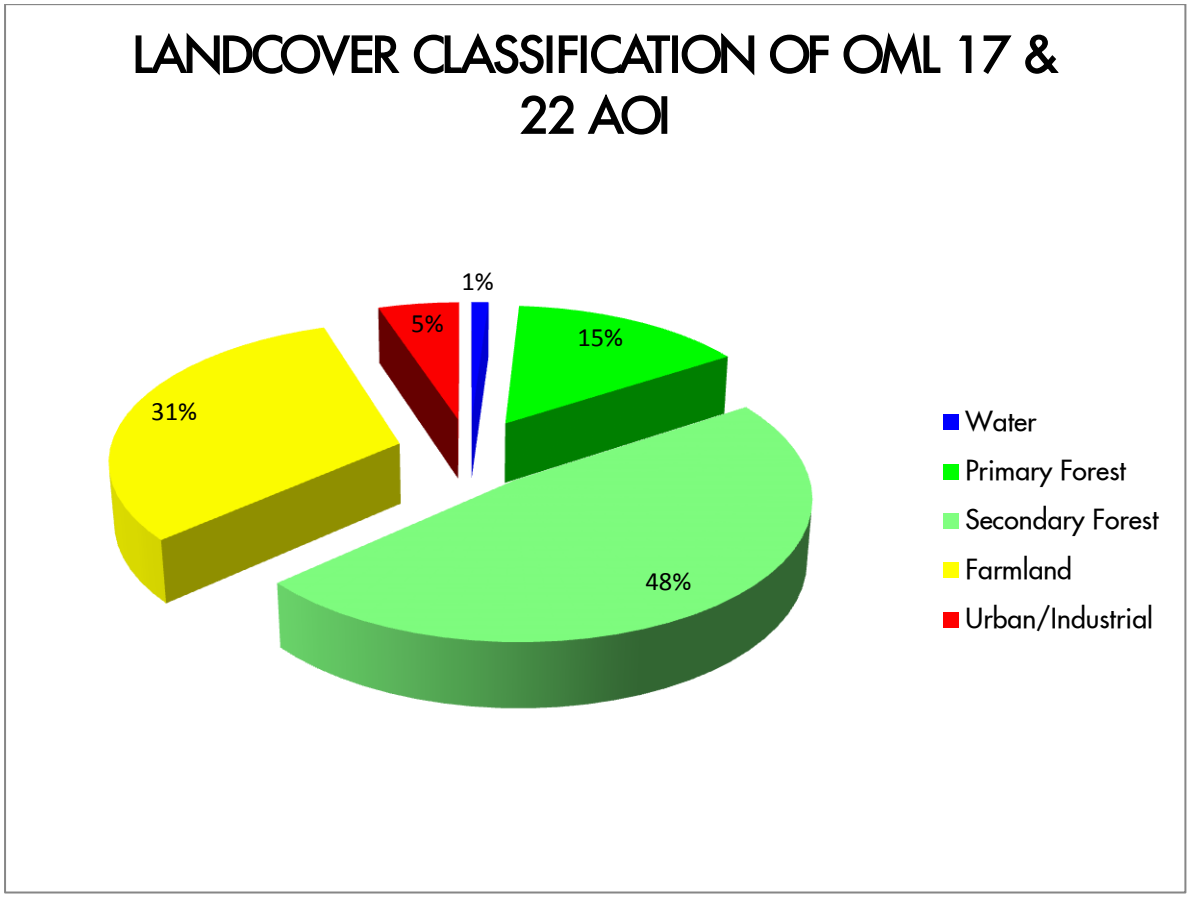

Fig 2: Pie chart of the Land cover types in January, 2016

5.0 FLOOD ANALYSIS MODEL

A flood analysis model was carried out integrating the Seismic DEM data and Land cover classification.

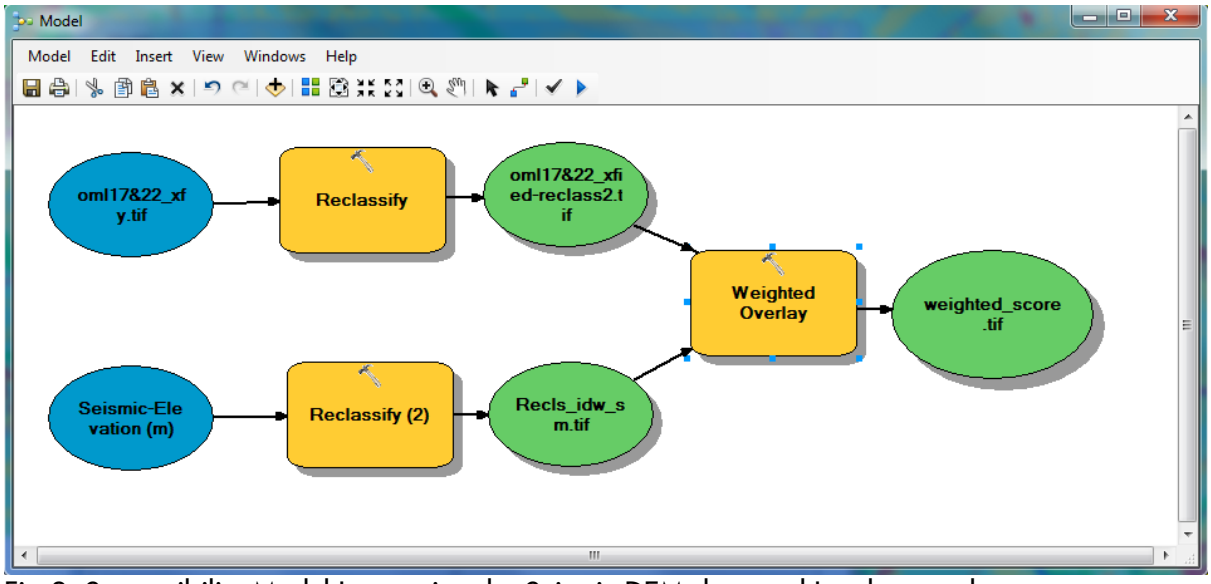

Fig 3: Susceptibility Model Integrating the Seismic DEM data and Land cover classes

Seismic DEM data was given 70% influence on the weighted score result while the Land cover classification was given 30% influence considering that the elevation has a more significant impact than land cover (*personal communication Iguniwari, 2015*).

The weighted overlay evaluation scale was varied on a scale of 1 to 5 by1 (where 1 is the highest elevation and 5 the lowest). The susceptibility was ranked between 5 - 1, where 5 is the highest susceptibility to flooding and 1 the lowest susceptibility.

Land cover was ranked on a scale of 1 to 5 relating on the susceptibility level of each of the classes based on visual interpretation thereby analysing each land cover class within the flood risk affected areas. From the analysis, Water bodies were ranked 5 as location at close proximity to water are more likely to be flooded, while Primary Forest, Secondary Forest, Farmland and Urban/Industrial were ranked 4, 3, 2 and 1 respectively.

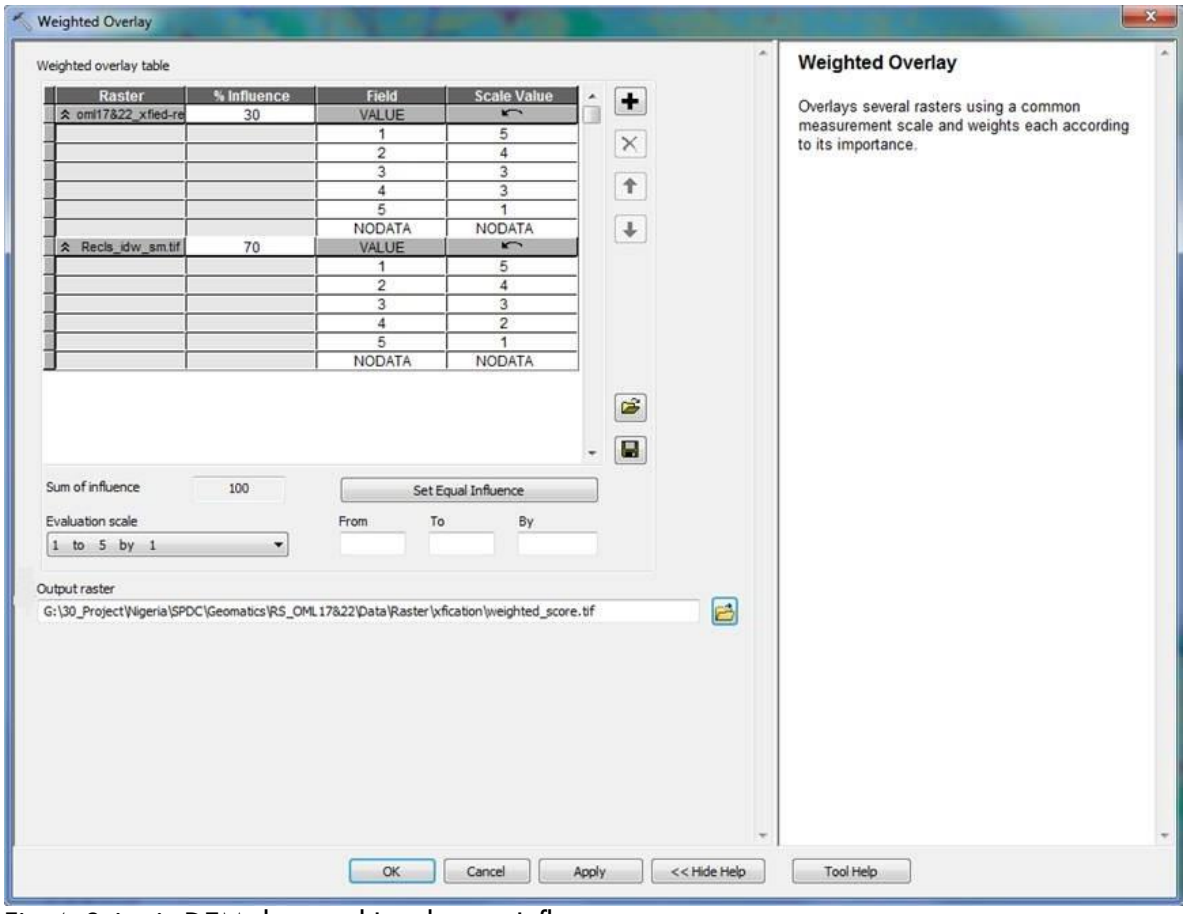

Fig 4: Seismic DEM data and Land cover Influence

6.0 MODEL VALIDATION

The flood susceptibility model was validated using the high water mark of 8.51m above mean sea level based on the 2012 SPDC High Water Mark record thereby reviewing the model prediction in relation to the 2012 Nun River water level monitoring data (please, see map no. EP201608214123005 for reference).

7.0 CONCLUSIONS

The susceptibility model map classes 4 and 5 (i.e. most areas at risk to flooding) align mostly with the 2012 High Water Mark record from the 2012 Nun River water level monitoring data. However, certain areas showed discrepancies. Hence, a composite map that incorporates areas of alignment between the susceptibility model analysis and the 2012 empirical flood data, area at high risk of flooding identified by model only, and the 2012 High Water Mark data indicating areas that are below/above the water mark were generated and highlighted as the maximum flood risk area within the OML 17/22 Seismic project area.

Appendix I
A0/A1. List of Hardcopy Maps accompanying this report

1.) EP201608214123001: January, 2016 Land cover map of study area
2.) EP201608214123002: Seismic DEM Data
3.) EP201608214123004: Map Showing Flood Risk Model and Flood Risk Model With Overlay of 2012 Empirical Flood Data
4.) EP201608214123005: Map Showing Areas That Are Below/Above the 2012 High Water Mark and Maximum Flood Risk Model Areas